**CSCE 222: Discrete Structures - Final Exam Reference Guide (Rosen 9th Ed.)**

## FUNCTIONS & RELATIONS

**diff:** a relation maps any domain element to codomain, a function has 1 output per input
**imags vs range vs codomain** Image is the mapping of a set, range is like the image, codomain is a set of possible values. preimage could be a set
**axioms** all inputs specified in domain must map to exactly one element of the codomain
**F=G** $\iff F(x) = G(x) \forall x \in X$
**1:1** $\iff \forall x1, x2 \in X, (F(x_1) = F(x_2)) \to (x_1 = x_2)$
**¬injective**: $\exists x_1, x_2 \in X, (F(x_1) = F(x_2)) \wedge (x_1 \neq x_2)$
**onto** $\iff \forall y \in Y, \exists x \in X | F(x) = y$
**¬surjective** $\exists y \in Y | \forall x \in X, F(x) \neq y$
Given bijective y=F(x), there exists $x = F^{-1}(y)$

## ALGORITHMS & COMPLEXITY

**Input:** Input values are of a specified set(type)
**Output:** Outputs are the solution(of a return type)
**Definitness:** Must define steps for the algorithm
**Correctness:** correct output for input set values
**Finiteness:** Finite steps return output for any input
**Effectiveness:** Each steps takes finite time
**Generality:** Works for all problems of desired form

**Upper bounds** $O(n!) > O(2^n) > O(n^2) > O(nlogn) > O(n) > O(logn) > O(1)$

| | | | |
|---|---|---|---|
| $O(1)$ | Array Access | $O(n^2)$ | Insertion Sort |
| $O(\log n)$ | Binary Search | $O(b^n)$ | Power Set |
| $O(n)$ | Linear Search | $O(n!)$ | Permutations |

$\mathbf{f(x) = O(g(x))} |f(x)| \leq C|g(x)|$ whenever $x > k$
find $C|g(x)|$: add $|c_i|$ terms to the largest term
$f_1(x) is O(g(x)) and f_2(x) is O(g_2(x))$:
$(f_1 + f_2)(x)$ is $O(max(|g_1(x)|, |g_2(x)|))$
$(f_1 \cdot f_2)(x)$ is $O(g_1(x) \cdot g_2(x))$
Measure the time complexity as the number of **operations** executed: while $i < n \wedge x \neq a_i \Rightarrow 3(n+1)$
**Lower Bounds:**
$f(x) = \Omega(g(x)) \iff f(x) \geq C \cdot g(x)$ for all x > k
find $C|g(x)|$ : add $c_i < 0$ terms to the largest term
**Tight Bounds:**
$f(x) = \Theta(g(x)) \iff f(x)$ is O(g(x)) and $\Omega(g(x))$ for all x > k
find $C|g(x)|$ : add $c_i < 0$ terms to the largest term

## SEQUENCES & SUMS

A sequence is either $\{a_i, \cdots, a_j\}$ or $\{a_n\}_{n \geq k}$
**Arithmetic Progression** a, a+d, a+ 2d, ..., a+nd
**Geometric Progression** $a, ar, ar^2, ..., ar^n$
$S_n = \sum_{n=0}^{n} ar^n = \frac{a(r^n-1)}{r-1}$
**Reccurrent Relation** $a_n$ in terms of previous terms
Plug in k-i to $a_n$ and solve for $a_n$ in terms of $a_{k-i}$
$a_n = n!$ is a **solution** to the recurrent relation $a \cdot n_{n-1}$
($a_n$ is in terms of n)
$\sum_{k=m}^{n} a_k + \sum_{k=m}^{n} b_k = \sum_{k=m}^{n}(a_k + b_k)$
$\prod_{k=m}^{n} a_k \cdot \prod_{k=m}^{n} b_k = \prod_{k=m}^{n}(a_k \cdot b_k)$
**Change of Variables** $\sum_{k=1}^{3}(k-1) = \sum_{j=0}^{2} j$

## INDUCTION & RECURSION

Recursions needs a stopping case(s), and to get closer to the stopping case.
(Weak, Strong, Inequality, Divisibility, Structural) Induction, check the proofs.
Set A = B $\iff (A \subseteq B) \wedge (B \subseteq A)$

## COUNTING

$T_1 = n_1, T_2 = n_2$
**Product Rule:** $T_1 \wedge T_2 = n_1 \times n_2 \Rightarrow 7(T_i) = n_i^7$
**For Sets:** $|A_1 \times A_2 \times \ldots \times A_m| = |A_1| \times |A_2| \times \ldots \times |A_m|$
**Sum Rule:** $T_1 \vee T_2 = n_1 + n_2$
**Disjoint S:** $|A_1 \cup A_2 \cup ... \cup A_m| = |A_1| + |A_2| + ... + |A_m|$
**Subtraction Rule:** $T_1 \vee T_2 = n_1 + n_2 - (T_1 \cap T_2)$
**Non disjoint sets:** $A_1 \cup A_2 = |A_1| + |A_2| - |A_1 \cap A_2|$
**Division Rule:** Unique Ways $= \frac{n}{d}$
**For Sets:** If a set $A$ is split into $n$ equal groups of size $d$, $n = \frac{|A|}{d}$
**Ex:** n seats at round table, Circular Ways $= \frac{n!}{n}$
**r-Permutations:** set of r ordered arrangements
**r-Combinations:** set of r unordered arrangements
P(n,r) $= \frac{n!}{(n-r)!}$ C(n,r) $= \binom{n}{r} = \frac{n!}{r!(n-r)!}$
P(n,r) with repetitions allowed $= n^r$
C(n,r) with repetitions (bins) allowed $= C(n-1+r, r)$
C order doesnt matter, therefore $C(n,r) = C(n, n-r)$
Binomial Theorem: $(x+y)^n = \sum_{j=0}^{n} \binom{n}{j} x^{n-j} y^j$
$\sum_{k=0}^{n} \binom{n}{k} = (1+1)^n = 2^n$
$\sum_{k=0}^{n} (-1)^k \binom{n}{k} = (-1+1)^n = 0$

## SOLVING RECURRENCES

Solve using preceeding elements
**Linear Homogenous recurrence relation:** $a_n = c_1 a_{n-1} + c_2 a_{n-2} + ... + c_k a_{n-k}$
**Linear:** RHS is a sum of previous terms[1] multiplied by some coeficient
**Homogeneous:** all RHS terms are multiples of previous terms$_{n-i}$
**Constant Coefficients:** all coefficients are constant, $\neq 0$
**Degree:** The number of previous terms in sequence
Degree determines NO. initial conditions $(a_0 = C_0), ..., (a_{k-1} = C_{k-1})$
**Characteristic Equations and Roots:**
$a_n = c_1 a_{n-1} + c_2 a_{n-2} + ... + c_k a_{n-k} \quad , \quad a_n = r^n \Rightarrow$
Where the solutions are the roots: $r^k - c_1 r^{k-1} - ... - c_{k-1} r - c_k = 0$
Solution $= (r_1, ..., r_k)^T \to a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$, $\alpha$ solved using initial condition
**Divide&Conquer Alg.time cmplx:** $f(n) = a \cdot f(\frac{n}{b}) + g(n)$, g(n) not constant?:
**Master Theorem Shorthand:**

$$f(n) = a \cdot f(n/b) + c \cdot n^d \implies \begin{cases} O(n^d) & a < b^d \\ O(n^d \log n) & a = b^d \\ O(n^{\log_b a}) & a > b^d \end{cases}$$

$$\begin{aligned} a &= \text{number of subproblems} \\ n/b &= \text{size of each subproblem} \\ c &= \text{constant factor} \\ n &= \text{input size} \\ d &= \text{exponent of overhead} \\ n^d &= \text{extra work (calls)} \end{aligned}$$

## RELATIONS

$R \subseteq A \times B \implies (aRb \iff (a,b) \in R)$, not all (a,b) are in R

- **Reflexive:** $\forall a \in A, aRa$
- **Symmetric:** $aRb \implies bRa$
- **Antisymmetric:** $(aRb \wedge bRa) \implies a = b$ *if both R work, must be reflexive*
- **Transitive:** $(aRb \wedge bRc) \implies aRc$
- Vacuously True: $P \implies Q$ is true if $P$ is always false. *(premise never met)*

you may use substitution to prove a=b or contradict
**Composition** $(S \circ R)$**:** $\{(a,c) \mid \exists b \in B, (a,b) \in R \wedge (b,c) \in S\}$
Using $S \circ R$ if a is parent of b, b is parent of c $\to$ a is grandparent of c in $S \circ R$
R on set A is an **equivalence** (like row equiv) if reflexive, symmetric, transitive: $(a,b) \in R \to a \sim b$

- **Powers** $(R^n)$**:** $R^1 = R, R^{n+1} = R^n \circ R$ *(path of length n)*
- **Equivalence Relation** $(\sim)$**:** A relation that is Refl. + Symm. + Trans.
- **Equivalence Class** $[x]$: $\{y \in S \mid x \sim y\}$ *(The set that act like x)*
- **Row Equivalence** $(A \sim B)$**:** An equivalence relation where $A, B$ share the same RREF.

**Directed Graph:** $G = (V, E)$ where $E \subseteq V \times V$ *(V=nodes/verticies, E=edges)*
**Loop:** $(a,a) \in E$ for some $a \in V$

## PROOFS

**Contradiction:** $P \to Q$
**Proof:** Assume that the hypothesis $P$ is true and that the conclusion $Q$ is false. [Insert logical steps here to derive a contradiction, such as $r \wedge \neg r$]. This is a contradiction. Therefore, our assumption that $Q$ is false must be incorrect, and $Q$ must be true.
**Induction:** $P(n)$ is true for all integers $n \geq a$.
**Proof:** Let $P(n)$ be the proposition [State P(n)].

- **Basis Step:** We show that $P(a)$ holds. [Verify the statement for $n = a$].
- **Inductive Step:** Assume the inductive hypothesis that $P(k)$ is true for an arbitrary integer $k \geq a$. That is, [State $P(k)$].
  We must show that $P(k+1)$ is true [State $P(k+1)$].

$$\begin{aligned} \text{RHS}(k+1) &= \underbrace{\text{General Terms up to } k}_{RHS(k)} + \text{Term}(k+1) \\ &= \text{RHS}(k) + \text{Term}(k+1) \\ &= \ldots \text{ [Algebraic Simplification] } \ldots \\ &\Rightarrow \text{RHS}(k+1) = LHS(k) + (k+1) \end{aligned}$$

$\therefore P(k) \to P(k+1) holds$
$\therefore$ [State $P(n)$] holds for all $n \geq a$ by mat. indu.

**Induction Inequality**: **Theorem:** Let $P(n)$ be the statement $[LHS(n)] < [RHS(n)]$ for $n \geq [n_0]$. **Proof:**

- **Basis Step:** For $n = [n_0]$, we have:

$$[LHS(n_0)] = [\text{value}] \quad \text{and} \quad [RHS(n_0)] = [\text{value}]$$

  Since [value] < [value], $P([n_0])$ is true.
- **Inductive Step:** Assume $P(k)$ is true for an arbitrary integer $k \geq [n_0]$. That is, assume:

$$[LHS(k)] < [RHS(k)] \quad \text{(Inductive Hypothesis)}$$

  We must show that $P(k+1)$ is true: $[LHS(k+1)] < [RHS(k+1)]$.
  Starting with the left-hand side:

$$\begin{aligned} [LHS(k+1)] &= [\text{Manipulate to find } LHS(k)] \\ &< [\text{Substitute } RHS(k) \text{ using Hypothesis}] \\ &< [\text{Logical "Bridge" Step like 2<(k+1)}] \\ &= [RHS(k+1)] \end{aligned}$$

$P(n)$ is true for all $n \geq [n_0]$ by mat. induction
**Strong Induction:** Multiple basis&more hypotheses 1. to prove hold 2. to use in inductive step